

Discrete Mathematical Structures			
Course Code	1BCSE301	CIE Marks	50
Course Type (Theory/Practical/Integrated)	Theory	SEE Marks	50
		Total Marks	100
Teaching Hours/Week (L:T:P)	4:0:0	SEE	3 Hours
Total Hours	50	Credits	04
Course Learning Objectives: 1. Use propositional logic in knowledge representation. 2. Apply principles of counting in computer applications 3. Find relation between different sets 4. Apply recurrence relations in real life problems 5. Use graph theory in computer applications			
Module-1:		10 hours	
Fundamentals of Logic: Basic Connectives and Truth Tables, Logical Equivalence: The laws of logic, Rules of inference. Open Statement, Quantifiers, Proof Methods Self Study Topic: Rule of Inference involving Quantifiers			
Module-2:		10 hours	
Combinatorics and Counting Principles Method of Mathematical Induction, fundamental principles of counting including the Rule of Sum and Rule of Product, permutations, combinations, and the multinomial Theorem Self Study Topic: Combination with Repetition			
Module-3:		10 hours	
Relations and Functions: Cartesian Products and Relations, Properties of relations, Equivalence Relations and Partitions. Functions, Function composition and Inverse function Pigeon Hole Principle Self Study Topic: Inverse of a function			
Module-4:		10 hours	
Enumeration and Generating Functions: Inclusion-exclusion principle, rook polynomials. First order linear recurrence relation, Second order linear homogeneous recurrence relations with constant coefficients Self Study Topic: Application of Rook Polynomial			
Module-5:		10 hours	
Graph Theory Graphs and sub graphs, Graph Isomorphism, Vertex degree, Planar Graphs, Graph Coloring, Trees and Sorting, and Prefix Codes. Self Study Topic: Dijkstra's shortest Path algorithms			

Course Outcomes: At the end of the course the student will be able to:	
1BCSE301.1	Apply knowledge of propositional logic in truth verification
1BCSE301.2	Demonstrate the use of principles of counting in computer science
1BCSE301.3	Recognize relations in real life applications
1BCSE301.4	Find applications of equivalence relation
1BCSE301.5	Apply inclusion principle, rook polynomial in computer applications
1BCSE301.6	Use graph theory in computer science

Sl.	Title of the Book	Name of the	Name of the	Edition and
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No		Author/s	Publisher	Year
Textbooks				
1	Discrete and Combinatorial Mathematics	Ralph P. Grimaldi and B V Ramana	Pearson Education, Asia,	5 th Edition – 2017
2	A first look at Graph Theory	John Clark and D. A. Holton	World Scientific Publishers	2 nd Edition 2011
Reference Books				
1	Discrete Mathematical Structures with Applications to Computer Science	J.P. Tremblay and R. Manohar	Tata – McGraw Hill Publications	1 st Edition ,2017
2	Discrete Mathematics and its Applications	Kenneth H. Rosen	Tata – McGraw Hill Publications	7 th Edition, 2017
3	Topics in Algebra	I N Herstein	Wiley Eastern Limited	2 nd Edition, 1993

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/111106052>- Introduction to Logic
- <https://nptel.ac.in/courses/106108227> - Counting Principles
- <https://nptel.ac.in/courses/106106183> -Discrete Mathematics
- <https://nptel.ac.in/courses/111106050> -Graph Theory

Course Articulation Matrix

Course Outcomes (COs)	Program Outcomes (POs)										
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
1BCSE301.1	2		1								
1BCSE301.2		2									1
1BCSE301.3	2	1									
1BCSE301.4		2									1
1BCSE301.5	2										1
1BCSE301.6	1	2									

1: Low 2: Medium 3: High